

Overview

Non-terrestrial networks (NTNs) are rapidly emerging as critical components of the nation's communications infrastructure, including low-Earth orbit satellite constellations, high-altitude platforms, and integrated space-air-ground communication systems. These systems provide global connectivity, positioning, navigation, and timing services, support critical infrastructure sectors, and defense applications. The rapid deployment of these large NTN architectures has elevated the security and resilience of these systems to an issue of strategic national importance. However, NTNs operate over shared wireless spectrum and tightly integrate space, air, and ground segments, creating a complex and highly exposed attack surface. Their reliance on long distance radio links and distributed network infrastructure makes them vulnerable to a wide range of threats, including smart jamming, spoofing, rogue network elements, electronic warfare techniques, infrastructure attacks, protocol vulnerabilities, and privacy breaches. Existing security mechanisms are insufficient to address these emerging threats (especially for a post-quantum world) as they are largely based on conventional encryption and authentication techniques. Consequently, this workshop will bring together leaders from academia, industry, and government to identify the most pressing security challenges facing current and future NTNs, developing a coordinated roadmap for the most imperative research issues.

Intellectual merit

Securing NTNs requires an interdisciplinary approach that spans wireless communications, network and systems security, cryptography, formal methods, artificial intelligence, resilience engineering, and space systems. As NTNs become an essential component of future communications critical infrastructure, their integration of space, air, and ground segments introduces novel attack surfaces and security challenges that remain poorly understood. The workshop will identify fundamental scientific challenges, synthesize ongoing research efforts, and establish new directions for interdisciplinary collaboration. Through focused discussions on signal-based attacks, infrastructure vulnerabilities, system-level weaknesses, and data and privacy risks, the workshop will generate a community-driven roadmap that advances the scientific foundations of NTN security and guides future research investments in this strategically important area.

Broader Impacts

The security and resilience of NTNs are critical to a broad range of stakeholders, including government agencies, commercial satellite operators, telecommunications providers, critical infrastructure operators, defense organizations, and the general public. As NTNs increasingly provide global connectivity, positioning, navigation, and timing services, and support essential sectors such as energy, transportation, and emergency response, vulnerabilities in these systems have the potential to create far-reaching economic and societal consequences. This workshop will provide a unique forum for researchers, industry practitioners, standards experts, and policymakers to collaborate on the security challenges facing current and future NTN deployments. The resulting research agenda is expected to inform federal agencies, industry stakeholders, and standards organizations as they develop strategies, architectures, and policies for securing future space-based communications infrastructure. In the longer term, the workshop will foster new interdisciplinary research collaborations, increase awareness of emerging security threats to NTNs, and help establish a community of researchers equipped to address the scientific and technological challenges associated with securing future space, air, and ground communication systems.